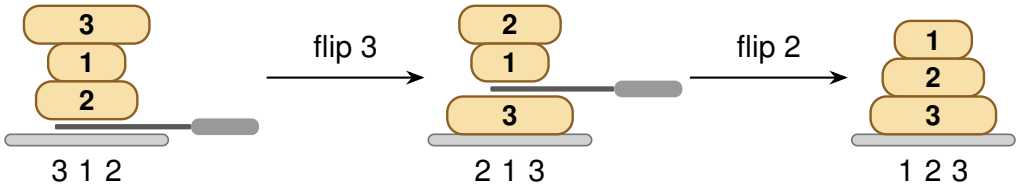
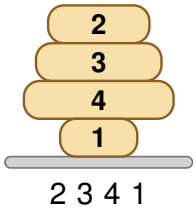
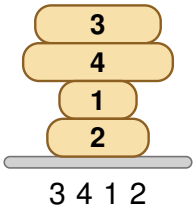
A stack is sorted when each pancake sits on a bigger one. To flip, slide the spatula under any pancake and turn that pancake and everything above it over together. Write a stack as its numbers from the top, and a flip as the number of pancakes it turns over.



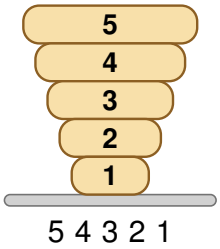
Problem 1: Sort each stack with as few flips as you can. Write down your flips.



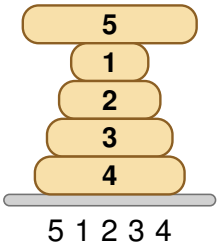
flips: _____



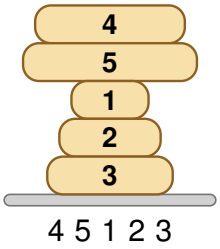
flips: _____



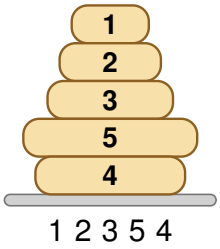
flips: _____



flips: _____



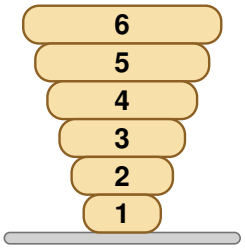
flips: _____



flips: _____

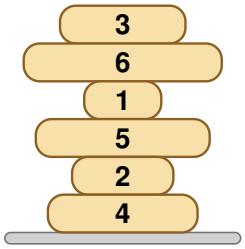
Problem 2: Which stack of pancakes 1, 2 and 3 needs the most flips? Explain why it cannot be sorted in fewer.

Problem 3: Sort each stack of six with as few flips as you can.



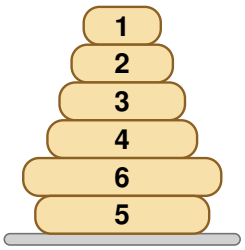
6 5 4 3 2 1

flips: _____



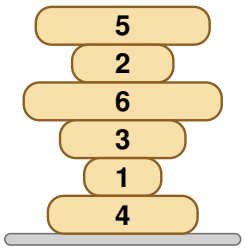
3 6 1 5 2 4

flips: _____



1 2 3 4 6 5

flips: _____



5 2 6 3 1 4

flips: _____

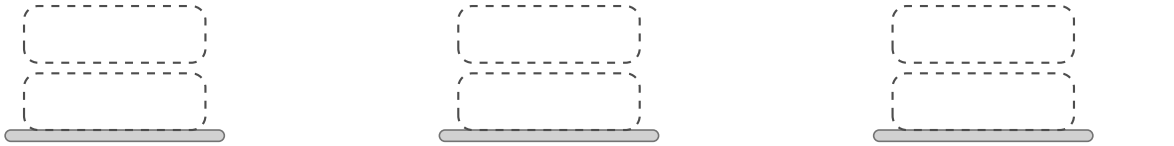
Problem 4: Find a way to sort any stack of six pancakes, however mixed up it is. What is the most flips your way can need? What about a stack of ten pancakes?

Problem 5: Find all the stacks of pancakes 1, 2, 3 and 4 that cannot be sorted in three flips or fewer.

Problem 6: Now each pancake has a dark side, which must end facing down. Sort each stack with as few flips as you can.

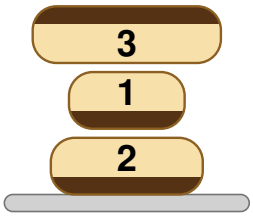
flips: _____ flips: _____ flips: _____ flips: _____

Problem 7: Find the stack of pancakes 1 and 2 that needs the most flips. Draw it with its dark sides.

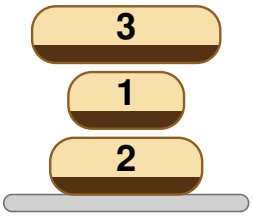


flips: _____ flips: _____ flips: _____

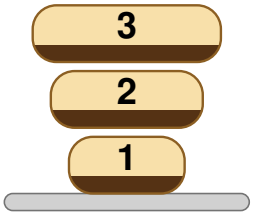
Problem 8: Sort each stack of three so that every dark side ends facing down, with as few flips as you can.



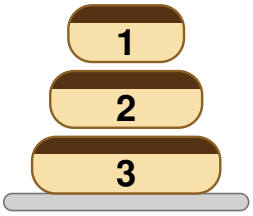
flips: _____



flips: _____



flips: _____



flips: _____